

Chapter 9

Arithmetic and Logic Unit

Arithmetic and Logic Unit

- Deals with the calculation and processing.

- Consists of 2 parts.

1. Arithmetic Unit:

-deals with the arithmetic operations of i/ps.

2. Logic Unit:

-deals with the logical operations of i/ps.

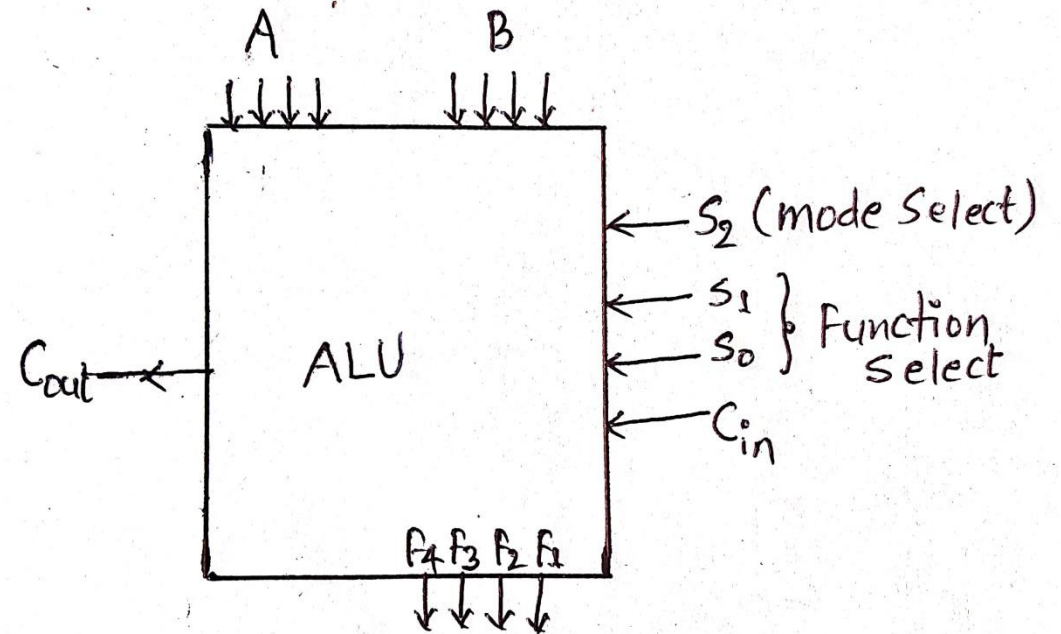


Fig: Block diagram of ALU

Eight different arithmetic operations

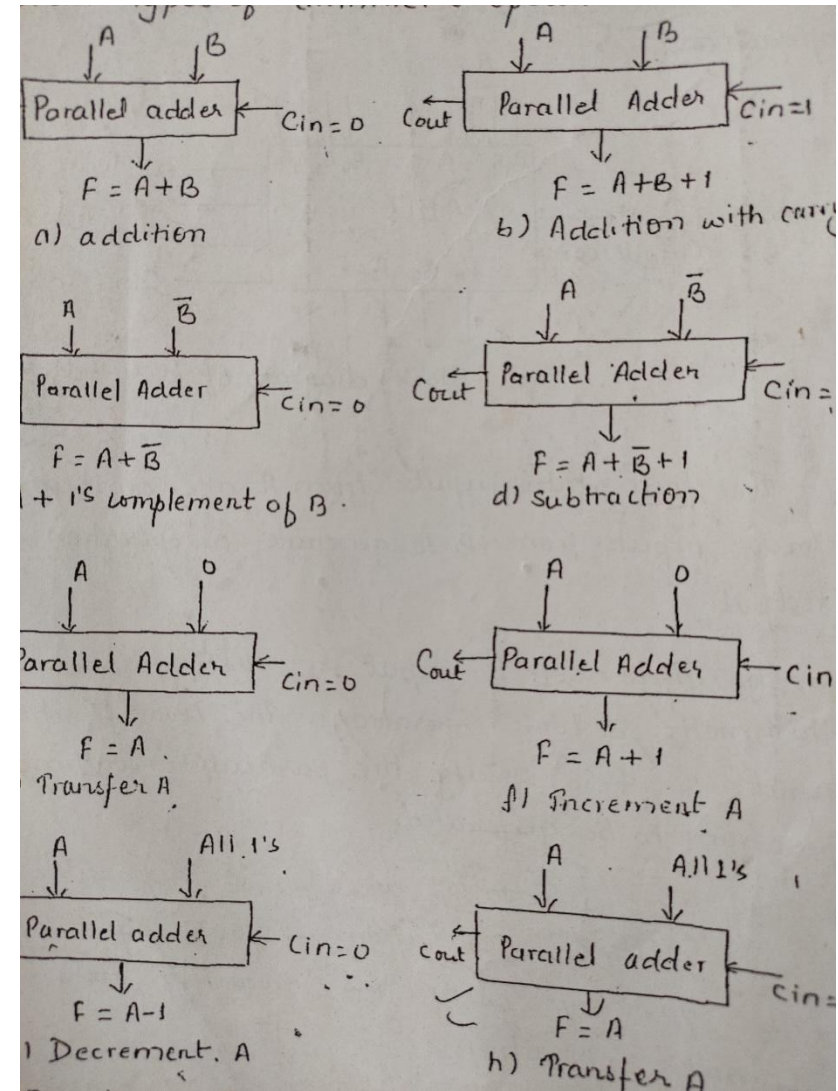
- The different operations are:

$$F = A + B \quad F = A + B + 1$$

$$F = A + B' \quad F = A + B' + 1$$

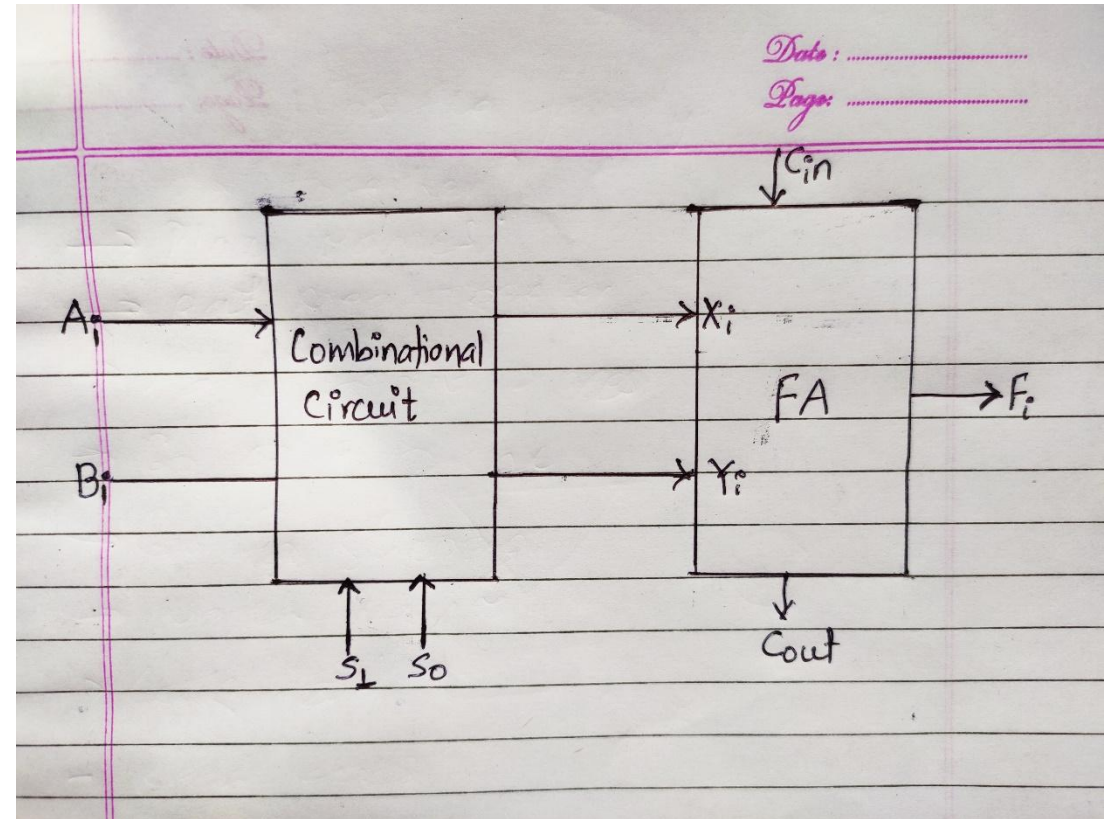
$$F = A + 0's \quad F = A + 0's + 1$$

$$F = A + 1's = A - 1 \quad F = A + 1's + 1 = A$$



Design of Arithmetic Circuit

- The block diagram of arithmetic circuit is as shown in figure along side.
- Its design include the design of combinational circuit before the parallel adder.
- The design starts with the specification or function table given.



Function table;

Function table:

S_1	S_0	C_{in}	F_i
0	0	0	$A_i \Rightarrow A_i^0 + \text{all } 0\text{'s}$
0	0	1	$A_i + 1 \Rightarrow A_i^0 + \text{all } 0\text{'s} + 1$
0	1	0	$A_i + B_i$
0	1	1	$A_i + B_i + 1$
1	0	0	$A_i + \overline{B_i}$
1	0	1	$A_i + \overline{B_i} + 1 \Rightarrow A_i - B_i$
1	1	0	$A + \text{all } 1\text{'s} \Rightarrow A_i^0 - 1$
1	1	1	$A + \text{all } 1\text{'s} + 1 \Rightarrow A_i$

Truthtable of the combinational logic:

Truthtable of the combinational circuit

S_1	S_0	A_i	B_i	X_i	Y_i	Remarks
0	0	0	0	0	0	$A_i + 0$'s
0	0	0	1	0	0	
0	0	1	0	1	0	
0	0	1	1	1	0	
0	1	0	0	0	0	$A_i + B_i$
0	1	0	1	0	1	
0	1	1	0	1	0	
0	1	1	1	1	1	
1	0	0	0	0	1	$A_i + \bar{B}_i$
1	0	0	1	0	0	
1	0	1	0	1	1	
1	0	1	1	1	0	
1	1	0	0	0	1	$A_i + 1$'s
1	1	0	1	0	1	
1	1	1	0	1	1	
1	1	1	1	1	1	

Using K-map,

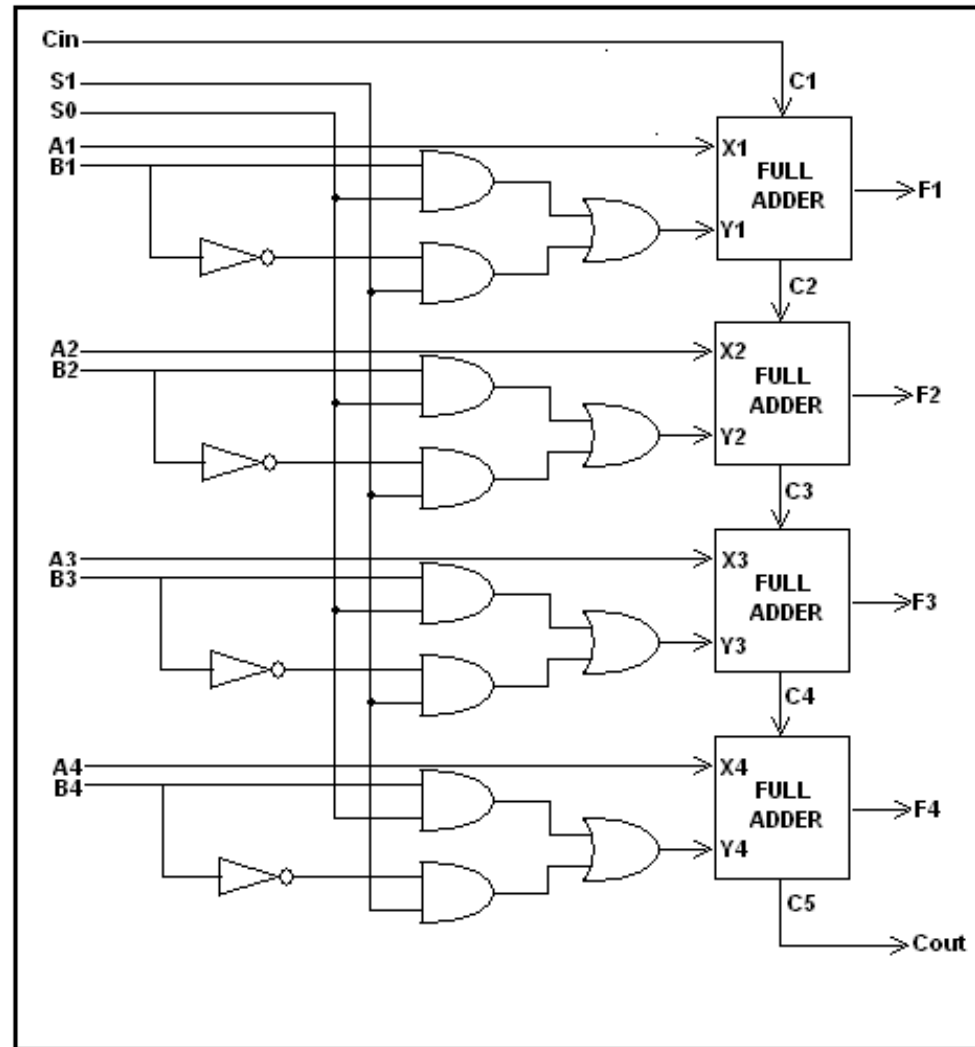
$$X_i = A_i$$

For Y_i ,

$S_1 \backslash S_0$	$A_i B_i$	00	01	10	11
00					
01			1	1	
11		1	1	1	1
10		1			1

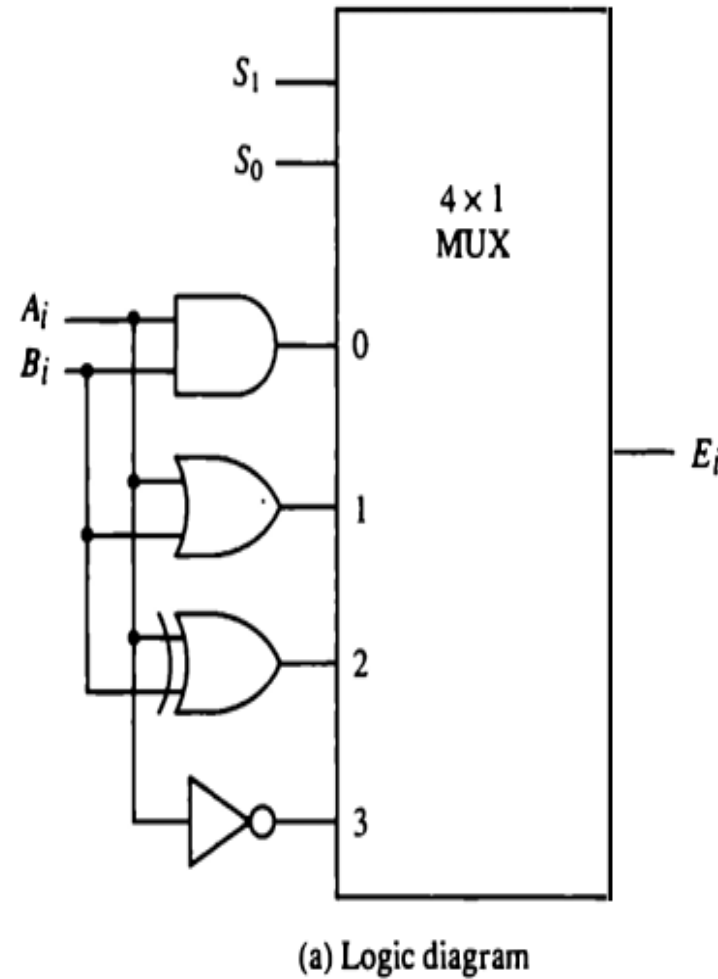
$$Y_i = S_1 \bar{B}_i + S_0 B_i$$

Circuit diagram of the 4 bit arithmetic circuit:



Design of Logic Unit:

- Select variables:
 S_2 -> mode select
 S_1, S_0 -> function select
- Let the logical operations be:
 OR, XOR, AND, NOT
- For design, let's take a 4×1 MUX



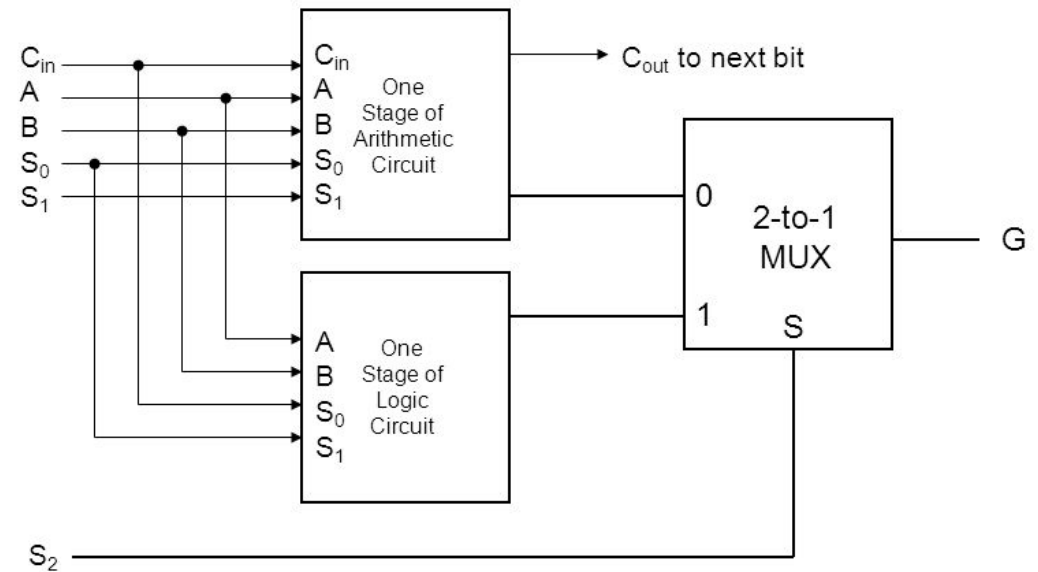
S_1	S_0	Output	Operation
0	0	$E = A \wedge B$	AND
0	1	$E = A \vee B$	OR
1	0	$E = A \oplus B$	XOR
1	1	$E = \bar{A}$	Complement

(b) Function table

Single stage ALU:

- Now combining both arithmetic and logic units we can draw or design ALU.
- The single stage ALU is as shown along side.

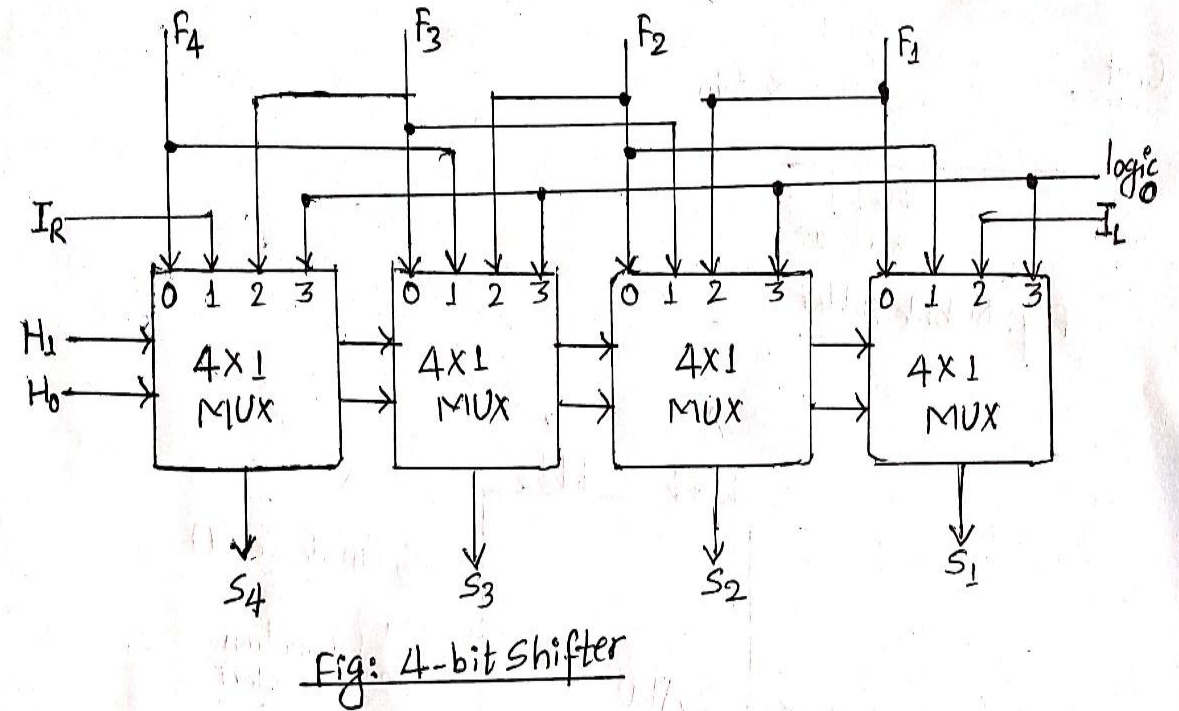
Putting Arithmetic and Logic Together



4-bit G Select becomes [S_2 S_1 S_0 C_{in}]

Design of Shifter:

- Transfers the o/p of ALU to the bus.
- It is a shift register with parallel load.
- The circuit diagram and the function table is as shown in the figure.



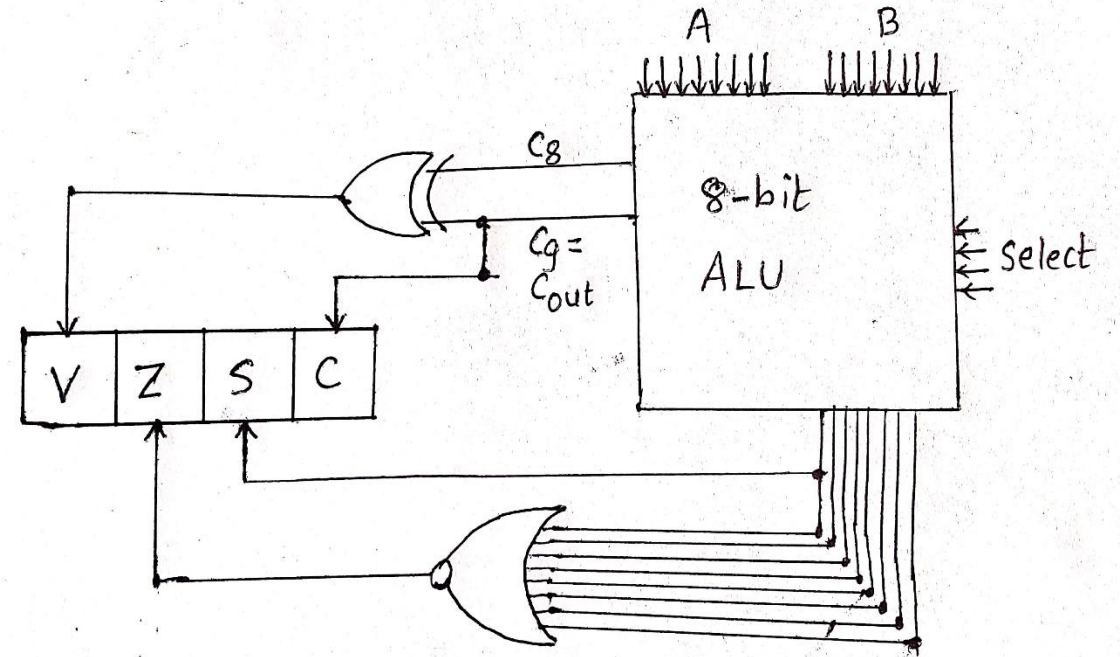
The function table of the 4 bit shifter is as:

Function table:

H_1	H_0	Operation	Function
0	0	$S \leftarrow F$	Transfer F to S
0	1	$S \leftarrow \text{shr } F$	Shift right F to S
1	0	$S \leftarrow \text{shl } F$	Shift left F to S
1	1	$S \leftarrow 0$	Transfer 0's into S

Status register

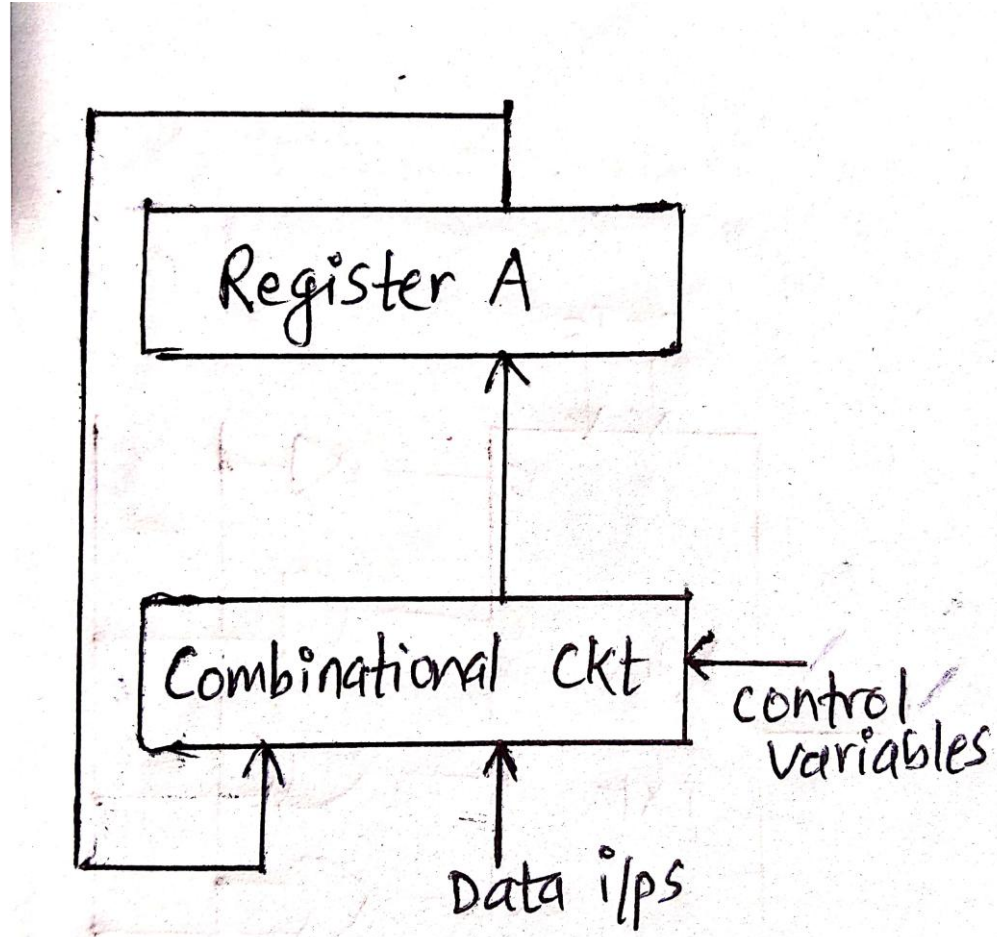
- Bit C (carry) is set if o/t carry of ALU is 1. It is cleared if the o/p is 0.
- Bit S (sign) is set if the highest order bit of the ALU o/p (the sign bit) is 1. It is cleared if the highest order bit is 0.
- Bit Z (zero) is set if o/p of ALU contains all 0's and cleared otherwise.
- Bit V (overflow) is set if the XOR of C8 and C9 is 1. This is the condition of overflow when the numbers are in sign 2's complement form.



Design of Accumulator

- Some processor units distinguish one register from all others and call it an accumulator register.
- Bidirectional register with parallel load connected to ALU.
- The accumulator and its associated logic constitute a sequential circuit.
- It consists of n -stages and n -flip flops.
- Register A is referred to as accumulator and sometime denoted by AC.
- It is a multifunction register that can perform all the microoperations.
- Microoperations of an AC depend on the particular processor

The block diagram and the different micro operations of an accumulator is as shown



Control variables	Microoperations	Name
P_1	$A \leftarrow A + B$	Add
P_2	$A \leftarrow 0$	clear
P_3	$A \leftarrow A'$	Complement
P_4	$A \leftarrow A \wedge B$	AND
P_5	$A \leftarrow A \vee B$	OR
P_6	$A \leftarrow A \oplus B$	XOR
P_7	$A \leftarrow \text{shr } A$	Shift right
P_8	$A \leftarrow \text{shl } A$	Shift left
P_9	$A \leftarrow A + 1$	Increment

Processor unit

- Selection variables in a processor unit control the micro-operations executed within the processor during a CP.
- Selection variables control the buses, ALU, shifter and the destination register.
- Processor unit consists of registers and status register. The o/p of register go thru 2 MUX to select the i/p of ALU.
- I/p data from external source are also selected by the same MUX.
- O/p of ALU goes to the registers or external destination thru shifter.
- There are 16 selection variables in the unit and their operations are controlled by a control word.

Processor unit continue...

- 16 bit control word, when applied, specifies a micro operation.
- Bit of A selects source register for the i/p to left side of ALU and Bits of B selects the source register for i/p to right side of ALU.
- D field selects the destination register.
- F field together with Cin select a function for ALU.
- H field selects the type of shift in the shifter.
- All 16 bit control words are stored in control memory and each word specify a micro operation.
- The sequence of control word is read from the memory, one at a time, to initiate a desired micro operation. This type of control operation is called micro programming.

The diagram of a processor unit is shown below:

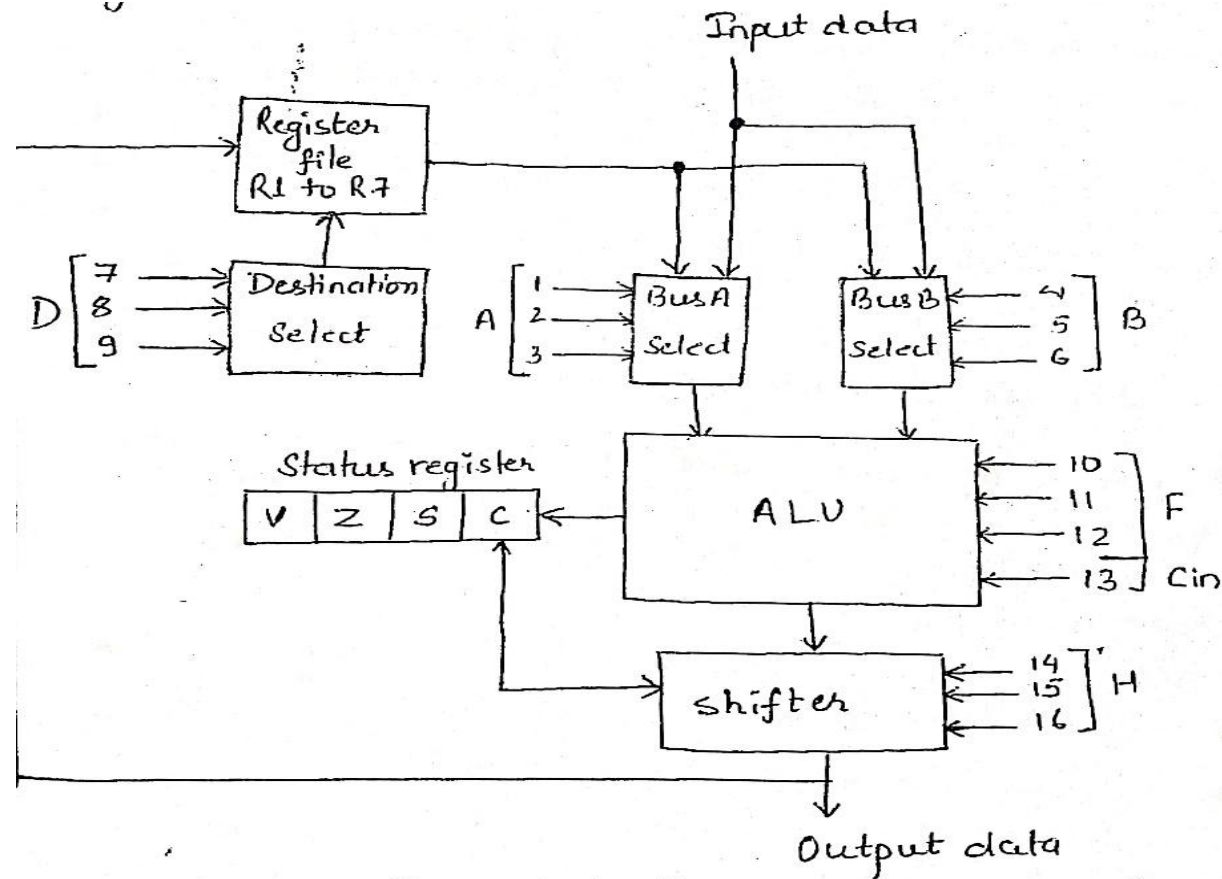


Fig: Block diagram

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
A	B	D	F	C _{in}	H										

Control word